

An Effective Lateral Transshipment Model for A Multi-Location Inventory Setting to Minimize Stockouts

A.G.K.C. Kumari

Dept. of Industrial Management
University of Kelaniya, Sri Lanka
komudyaz.7@gmail.com

A. N. Wijayanayake*

Dept. of Industrial Management
University of Kelaniya, Sri Lanka
anni@kln.ac.lk

D. H. H. Niwunhella

Dept. of Industrial Management
University of Kelaniya, Sri Lanka
hirunin@kln.ac.lk

Abstract - Managing inventory levels to ensure on-shelf availability of products is a challenge that retailers face on a daily basis. Even though it is desirable to have additional inventory to ensure the availability of products, it increases the inventory holding cost. Hence, retailers use lateral transshipment as a method to redistribute inventory from a location which has excess inventory to another outlet which faces / will face stockouts. This paper proposes a mathematical model to minimize the total cost through proactive lateral transshipment while reducing the stockouts, significantly. A multi-item, multi-location inventory system was considered, and a cost minimization model was developed based on the tradeoff between the potential gain and the transshipment cost. The model was implemented using Python programming language and validated using a real-world data set from one of the leading supermarket chains. The results from the model have shown that it can reduce the total cost and stockout occurrences significantly.

Keywords—inventory management, lateral transshipment, proactive transshipment, retail industry

I. INTRODUCTION

The main challenge a retail chain has to face is managing sufficient inventory levels in all stores of the chain. It has become more challenging with the increasing demands and expectations of customers due to factors like the expansion of e-commerce and intensified competition. The competition among retailers demands them to ensure the availability of the right product at the right place and right time [1]. Hence, to keep up with the competition, retailers thrive to ensure the availability of products. However, [2] states that the capability of a retail centre to satisfy the demand is not solely reflected by the stock position. Instead, the availability of the product (or a satisfactory substitute) within the acceptable delivery time of a customer determines the ability of the demand satisfaction of a particular retailer.

Efficient inventory management is a critical factor in ensuring product availability across all the stores of a retail chain. [3] point out that it is critical to effectively manage the inventory as it often accounts for a large proportion of the product cost. Hence, even though maintaining high inventory levels could ensure customer satisfaction through continuous product availability, high inventory levels lead to higher costs. Therefore to obviate the high inventory costs, it is critical to maintain minimum sustainable stock levels [4].

Companies use modern technology to predict their future sales, but mostly, those forecasts are based on historical sales data. [5] point out that the actual demand can significantly deviate from the predicted values based on multiple external factors (for example weather conditions), and it will lead to some stores to have excess inventory while other stores face stockout occurrences. In a situation like that, lateral transshipment provides an effective solution to address the mismatch. Lateral transshipment is defined as “sharing or rotating stocks in a multi-location inventory system at the same echelon in the supply chain” [4]. It can be executed either proactively, where stocks are redistributed periodically at a predetermined time or reactively, where stocks are redistributed in a way to meet the demand which cannot be fulfilled from available inventory.

Most of the literature on lateral transshipment has focused on the spare parts industry; specifically under reactive transshipment [5]. But with the wide application of technology and ERP systems, the readily available real-time data enable the application of proactive lateral transshipment as well [5]. However, the majority of the existing studies have focused on reactive transshipment [5][6]. Hence, relatively less attention has been given to proactive lateral transshipment. The complexity of the proactive lateral transshipment increases as it has to determine the quantity to transship, and the sourcing outlet as well. Hence, studies have primarily focused on a single-period or two-location inventory systems [5]. Hence, relatively less attention has been given to multi-location proactive lateral transshipment models. Further, [3] have identified that considering multiple items for the supply network should be a new development avenue in the lateral transshipment literature as it opens up several issues as well as possibilities.

This research is carried out to propose a new model to minimize the total cost while reducing the stockout occurrences through lateral transshipment in a supermarket chain. A multi-location, multi-item setting was considered for the study. The objective of this research is to propose a cost minimization model to determine whether transshipment should be carried out based on the trade-off between the potential gain and the cost of the lateral transshipment. Further, the model determines the source-destination pair for the transshipment as well as the amount to be transshipped.

The rest of the paper is organized as follows. Section 2 provides a detailed review of the

existing literature related to lateral transshipment. Section 3 describes the methodology followed in carrying out the research. Section 4 describes the data analysis and the results, followed by Section 5, which presents the conclusion of the research.

II. LITERATURE REVIEW

Use of lateral transshipments have been studied under different inventory settings and constraints. It has been utilized by many industries such as spare parts, automotive, apparel, furniture, and sporting goods [7]. The main classification of lateral transshipment literature is based on the timing of the transshipment. Mainly it is divided into proactive and reactive transshipment. Proactive transshipment is also known as preventive transshipment, and it allows to redistribute stocks at a predetermined point in time [8]. Since there is an opportunity to arrange the transshipment in a way to reduce handling cost, preventive transshipment is more effective for settings where handling cost is high [3]. Reactive transshipment is executed once the demand is realized. When the ability to instantaneously respond to customer demand is low and when the cost of transshipment is less than the inventory holding cost, reactive transshipment can be beneficial [3]. There is also another avenue in transshipment literature which combines proactive and reactive transshipment characteristics. For example, decision rules for a hybrid transshipment approach for a two-echelon inventory system was developed [4].

A. Reactive Lateral Transshipment

One of the first studies in lateral transshipment was done by [9] and, a centralized, single-echelon inventory system was studied under a periodic review policy with the cost minimization objective. Since then, many researchers have studied reactive lateral transshipment. The majority of transshipment literature falls under the reactive transshipment category [5][6]. The spare parts industry is the most studied industry under reactive transshipment [5]. For example, the study carried out by [10] considered one or two-echelon systems with nonnegligible transshipment lead time. The authors have proved the importance of including pipeline stock flexibility along with lateral transshipment in inventory-related decision-making.

Reference [11] have tested multiple transshipment policies and identified that all the policies give better results than the no-transshipment policy. They have considered four different pooling scenarios. However, as pooling can be done in other different ways based on the circumstances, the ability to generalize the model is doubtful. Some authors have also studied two-echelon inventory systems. For example, [12] have developed a Markovian model for a two-echelon inventory system consisting of a supply plant and several warehouses. One of the drawbacks of the study is that they have considered identical warehouses in terms of capacity and cost structure. Hence, it was proposed to extend this study to non-identical warehouses.

Differentiating from the above studies, the importance of initial stocking decision along with the transshipment of inventory, was studied by [13]. The study has proven that initial stocking decisions and transshipment decisions during

the season contribute to the profitability in different order magnitude. Specifically, for fast-moving and high-volume products, the initial stocking decision has proved to be a high contributor.

This paper differentiates from the studies mentioned above as a proactive lateral transshipment approach was considered for this study.

B. Proactive Lateral Transshipment

Proactive lateral transshipment has gained less attention in literature compared to reactive transshipment [14]. One of the earliest studies in proactive transshipment is [15], where a single period, multi-location setting was considered.

Reference [16] have studied a single period proactive transshipment model for a fashion retailer with deterministic demand. The authors have used a metaheuristic approach in solving the problem due to the high complexity caused by the operational constraints. Deterministic demand and the high number of operational constraints which makes the generalizability of the model doubtful, are some of the drawbacks of this model.

Some authors have considered multi-period proactive transshipment settings as well. Another avenue in transshipment literature is where lateral transshipment models are coupled with inventory routing problems. Those are called 'inventory routing problems with transshipment' (IRPT). [17] has introduced IRPT for a situation where vehicle routing and transshipment decisions have to be made simultaneously. Usually, those problems are considered nondeterministic polynomial hard (NP-hard) problems, and metaheuristic approaches are utilized in solving those problems. Extending the work done by [17], an inventory routing problem with planned transshipment for multi-period, multi-item and multiple vehicle system was studied by [18].

Incorporating environmental objective functions along with economic objective functions have been another research avenue in transshipment literature. [19] have formulated a mixed integer problem for an inventory routing problem with a transshipment option with a green approach. They have considered a multi-product, multi-period, multiple capacitated vehicle setting.

Multi-location and multi-period proactive transshipment studies are rare in literature. This is due to the complexity of the problem. Researchers would mostly consider either single period or two-location inventory systems or both [5]. However, the work done by [20] is an exception for that. The authors have taken a dynamic programming approach for a multi-location inventory system, where in each period, dynamic decisions are made to ensure profit maximization in the remaining periods.

C. Hybrid Transshipment Approach

The concept of hybrid transshipment was first introduced by [21], where they introduced an enhanced reactive transshipment policy which transfers additional inventory to reduce future stockouts as well. [4] have developed a decision rule to make trade off decisions based on different cost elements to choose between back ordering and hybrid transshipment policy. Transshipment policy. A simulation

model for a two-echelon inventory model was developed by [22].

A multi-period hybrid transshipment model was studied by [23]. The model has followed the quasi-approach used by [21]. The authors have considered a multi-location, multi-item inventory system. [24] have developed a hybrid transshipment model to reduce the downtime of a spare parts distribution system by integrating reactive and proactive approaches.

The proposed study is different from the hybrid approaches mentioned above, as only a proactive lateral transshipment model is considered for our model.

Recent lateral transshipment related literature has been categorized and shown in Table I. Studies conducted between 2010 to 2020 were considered. The literature is categorized according to the type of transshipment (proactive, reactive or hybrid), considered time duration in the model (single period, multi-period), and the number of items (Single item, multiple items). The reference number of the literature is given in [#] as shown in the table below.

TABLE I: RECENT LATERAL TRANSHIPMENT LITERATURE SUMMARY

	Proactive transshipment		Reactive transshipment		Hybrid transshipment	
	Single period	Multi-period	Single period	Multi-period	Single period	Multi-period
Single item	[20]	[25]	[25]		[4]	
	[26]	[21]	[27]		[22]	
	[17]	[10]	[13]			
	[5]	[28]	[29]			
	[6]	[12]	[30]			
	[31]	[11]				
Multi-item	[32]					
	[16]	[19]	[33]	[34]		[23]
		[18]	[8]	[35]		[36]

According to the review of literature, less attention has been given to multi-location, multi-item, single-period proactive lateral transshipment models. Hence, to fill the gap in the literature, in this research a multi-item, multi-location proactive lateral transshipment model is developed. The model is able to identify the sourcing outlet and the number of items to be transshipped as well. The model will also only allow the transshipment if only a net gain can be earned through the transshipment. The transshipment decision is based on the cost of the transshipment and the possible gain that can be gained.

III. PROBLEM DESCRIPTION AND MODEL DEVELOPMENT

A. Problem Description

This research has considered a centralized inventory system with one central warehouse and multiple retail outlets (the word outlet and location are used interchangeably throughout). The time duration between two adjacent replenishments is known as the review cycle (T).

At the beginning of a review cycle, the outlets are replenished with the regular order to bring the stock level to the base stock level. Following the approach used by [20], equation 1 is used to define the base stock level (U_{pi}).

$$U_{pi} = \mu_{pi}T + k_{pi}\sigma_{pi}T \quad (1)$$

This includes the expected demand plus a safety stock of $k_{pi}\sigma_{pi}T$ items. k_{pi} is the safety stock factor [20]. μ_{pi} and σ_{pi} represents the mean value and standard deviation of demand for item p in location I, respectively. A simplified version of the model is shown in Fig. 1 for a single item.

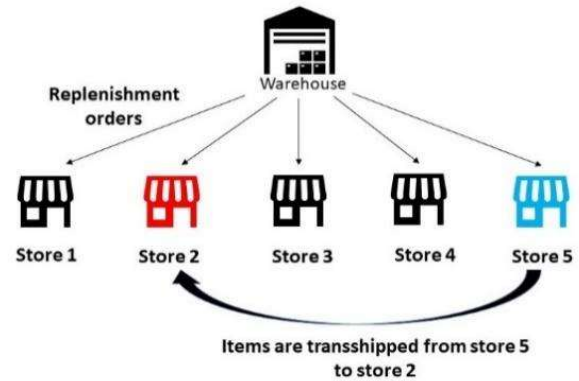


Fig. 1. The simplified version of the proposed model

In the simplified version of the model a single item is considered. After t times into the replenishment cycle, the outlet which is going to face a stockout is identified using an equation (Denoted as store 2 in Fig. 1). The inventory levels in the rest of the stores are checked and based on the highest gain the sourcing outlet will be selected. (Denoted as store 5 in Fig 2.)

B. Model Development

A mathematical model is developed with the objective of minimizing the total cost. The total cost includes the inventory holding cost, shortage cost and the transshipment cost. The assumptions of the mathematical model are given below.

Assumptions:

- A centralized inventory system with one central warehouse and multiple retailer outlets is considered.
- Demands are deterministic and independently distributed in each location for each product.
- The retail stores adopt periodic order policy. All the stores follow identical review periods.
- At the beginning of each review cycle, the replenishment happens in each outlet to bring the stocks to a base stock level.
- Only one order cycle is considered.
- Only one transshipment opportunity is allowed per review cycle.
- Unsatisfied demand is lost.
- Backordering is not allowed.
- The transshipment lead time is negligible.
- The same type of vehicles with unlimited capacity are used for the transshipment.

The notations used in the mathematical model are shown in Table II.

TABLE II. NOTATIONS AND THE DEFINITIONS OF THE MATHEMATICAL MODEL

Notation	Description
Indices/ Sets	
$i, j \in L$	Set of outlets ($i, j = 1, 2, \dots, n$)
p	Set of items ($p = 1, 2, \dots, m$)
T	Length of replenishment cycle ($t \in T$)
Input variables	
μ_{pi}	Mean value of demand for item p in location i
σ_{pi}	Standard deviation of demand for item p at location i
I_{pi}	Current inventory level of item p at location i
I_{pi}^+	Inventory level of item p at location i after transshipment
P_p	Selling Price of item p
C_p	Purchase cost of item p
H_p	Inventory holding cost per item p
L_p	Shortage quantity of item p after transshipment completed
d_{ij}	Distance between location i and location j
TC	Transportation cost per kilometer
G_{pij}	Gain from the transshipment of item p from store j to store i
k_{pi}	Reserve stock safety factor
Decision variables	
q_{pij}	The transshipment quantity of item p from store j to i
X_{ij}	Binary variable that determines whether a transshipment happens from j to i
N_{pi}	The target quantity to fulfil the demand of item p in store i for $(T-t)$ period

Objective function

$$\text{Minimize} \sum_{p \in P} \sum_{i \in L} (h_p I_{pi}^+ + d_{ij} TC X_{ij} + (P_p - C_p) L_p) \quad (2)$$

Subjected to:

$$N_{pi} = \mu_{pi}(T - t) + k_{pi} \sigma_{pi} \sqrt{(T - t)} \quad \forall i \in L, p \in P \quad (3)$$

$$q_{pji} \leq N_{pj} \quad \forall i, j \in L, p \in P, i \neq j \quad (4)$$

$$I_{pi}^+ = I_{pi} + q_{pji} X_{ij} \quad \forall i \in L, p \in P \quad (5)$$

$$G_{pij} \leq [(P_p - C_p) q_{pji}] - d_{ij} TC \quad \forall (i, j) \in L, p \in P \quad (6)$$

$$X_{ij} \in \{1, 0\} \quad (7)$$

- Objective is to minimize the total cost of the inventory model. Total cost includes the inventory holding cost, transshipment cost and the shortage cost.
- Defines the required target inventory level to fulfil the upcoming demand in $(T-t)$ time period.
- Store j may only transfer the excess inventory to store i . (The store j will not transfer the inventory which is required to fulfil their demand in $(T-t)$ time period.)
- Defines the stock levels after the transshipment.
- Transshipment between store j and store i may only be executed if the net profit of the transshipment is positive
- Binary variable to determine whether a transshipment happens between i and j .

Note—store i is considered as the store facing stockouts and store j is considered as the sourcing outlet.

IV. DATA ANALYSIS AND RESULTS

The mathematical model was implemented using Python 3.8.5 programming language in a Jupyter Notebook environment. To test the model a real-world dataset has been obtained from one of the leading supermarket chains in Sri Lanka. The dataset consists of data related to 3 products and 5 outlets. Therefore, here we consider $p = 1, 2, 3$ items and outlets $i = 1$ to 5.

The results obtained from the model execution is presented in Table III. The results are shown as a comparison between the baseline model (no transshipment) and the proposed model. Cost reduction obtained through the proposed model is also shown as a percentage.

TABLE III. THE RESULTS OBTAINED FROM THE MODEL EXECUTION

		Product 1	Product 2	Product 3
Costs without transshipment scenario (LKR)	Shortage Cost	510.53	129.57	148.56
	Transfer Cost	0.00	0.00	0.00
	Inventory Cost	413.09	856.10	95.65
	Total Cost	923.62	985.67	244.20
Costs with lateral transshipment scenario (LKR)	Shortage Cost	110.61	0.00	15.74
	Transfer Cost	193.50	112.50	135.00
	Inventory Cost	260.93	757.38	52.46
	Total Cost	565.04	869.88	203.20
Cost reduction as %	Shortage Cost	78.33%	100%	89.40%
	Transfer Cost	-	-	-
	Inventory Cost	36.83%	11.53%	45.15%
	Total Cost	38.82%	11.75%	16.79%

According to the results of the model, there is a significant cost reduction in the proposed scenario compared to the baseline scenario. Even though the company has to incur additional transfer cost, higher cost saving has been achieved through the reduction in shortage cost and inventory holding cost. The reduction in shortage cost also depicts that the out-

of-stock occurrences will be reduced, and hence the impact on the customer loss of goodwill is minimized.

V. CONCLUSION

Lateral transshipment has gained attention in the literature as well as among practitioners as an effective way of achieving sufficient inventory levels in the outlets of the same echelon in the supply chain. It could also be useful in reducing the stockout occurrences and thereby increase the customer satisfaction. In the retail industry, ensuring maximum on-shelf availability of products is a key performance indicator. Hence, lateral transshipment is known to be an effective solution for the mismatch of inventory levels in a retail chain.

Through the literature review, it was identified that proactive lateral transshipment had gained relatively less attention [14]. Most of the literature has focused on reactive lateral transshipment [5]. Moreover, there are only a few studies that have considered multi-item, multi-location proactive lateral transshipment. Hence, this paper proposes a mathematical model for multi-location, multi-item inventory systems to proactively make the lateral transshipment decision based on the trade-off decision while ensuring maximum gain. The proposed model can determine the sourcing outlet as well as the quantity to transship from the source. The transshipment decision is executed considering the transshipment cost and the cost of lost sales. This model also minimizes the total cost while proactively reducing stockouts.

The data from one of the leading supermarket chains were used to validate the model, and the results shows a total cost reduction of 23.93%, compared to the baseline scenario. Further, the reduction in cost of lost sales and inventory holding cost compared to the baseline scenario amounted to 83.98% and 21.55%, respectively. Hence, it proves that the utilization of the proposed model results in significant cost reduction and reduction in stockouts. Therefore, the model helps to increase customer satisfaction and improve the goodwill of the company through increased availability of products.

It should be noted that lead time for the transshipment was considered to be negligible for the ease of model formation. However, in real-world scenario, it might not be the case. Further, only one transshipment opportunity was allowed for a review cycle. But for the products with longer replenishment cycles, transshipment needs might occur multiple times per review cycle. Hence for future research, the model can be further developed to allow multiple transshipment opportunities per review cycle. Allowing a dynamic decision-making point instead of a predetermined time point for transshipment execution and relaxing the constraints like negligible lead times can further enhance the practicality of the model.

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